

Two exercises

Pierre-Yves Gaillard

This text is available at <https://github.com/Pierre-Yves-Gaillard/weil-beginners-exercises>.

This text has been written with help of Gemini and Qwen. Its purpose is to solve Exercises III.5 and III.6 in Weil's book [2]. Here are the statements (in a slightly rephrased form):

Exercise 1 (Exercise III.5 in [2]). *Let a and b be two relatively prime positive integers and n an integer $\geq ab$. Show that there are nonnegative integers x, y such that $n = ax + by$.*

Exercise 2 (Exercise III.6 in [2]). *Let $a_1, \dots, a_k$ be relatively prime positive integers and n a positive integer. Using Exercise III.5 and induction on k , show that, if n is large enough, then there are nonnegative integers $x_1, \dots, x_k$ such that $n = a_1x_1 + \dots + a_kx_k$.*

After solving Weil's exercises, we prove Proposition 2.1 in Gallier's text [1], which gives a stronger result.

The following lemma will be useful to solve the two exercises.

Lemma 3. *In the setting of Exercise III.5, let e, f be nonnegative integers. If $n \geq ab + ae + bf$, then there are integers x, y such that $n = ax + by$, $x \geq e$, $y \geq f$.*

Proof. Let $n \geq ab + ae + bf$. Since a and b are relatively prime, there are integers u, v such that $n = au + bv$, and any integer solution (x, y) to the equation $n = ax + by$ is of the form $(u + kb, v - ka)$ for some integer k . This gives $x = u + kb$, $y = v - ka$. It is enough to show that there is an integer solution k to the inequalities $u + kb \geq e$, $v - ka \geq f$. This can be rewritten as

$$\frac{e - u}{b} \leq k \leq \frac{v - f}{a}.$$

A sufficient condition for the existence of such an integer k is

$$\frac{v - f}{a} - \frac{e - u}{b} \geq 1.$$

This condition is satisfied because we have

$$\frac{v - f}{a} - \frac{e - u}{b} = \frac{au + bv - (ae + bf)}{ab} = \frac{n - (ae + bf)}{ab} \geq \frac{ab}{ab} = 1.$$

□

Exercise III.5 is solved by setting $e = f = 0$ in Lemma 3. We now solve Exercise III.6 by induction on k . We may assume that $k \geq 3$ and that the result holds for $k - 1$. Let d be the gcd of $a_1, \dots, a_{k-1}$. In particular, d and a_k are relatively prime. Let $a'_i = a_i/d$ for $1 \leq i \leq k - 1$. The integers $a'_1, \dots, a'_{k-1}$ are relatively prime. By the induction hypothesis, there is a positive integer n_0 such that any integer $u \geq n_0$ can be written as $u = a'_1x_1 + \dots + a'_{k-1}x_{k-1}$ for some nonnegative integers $x_1, \dots, x_{k-1}$. Multiplying by d , this implies that $du = a_1x_1 + \dots + a_{k-1}x_{k-1}$. Let $n \geq da_k + dn_0$. By Lemma 3 applied to the relatively prime integers d and a_k , there are integers u, x_k such that $n = du + a_kx_k$, $u \geq n_0$, $x_k \geq 0$. By the induction hypothesis, there are nonnegative integers $x_1, \dots, x_{k-1}$ such that $du = a_1x_1 + \dots + a_{k-1}x_{k-1}$. This yields $n = a_1x_1 + \dots + a_kx_k$ with $x_1, \dots, x_k \geq 0$, as required.

We now prove Proposition 2.1 in Gallier's text [1].

Proposition 4. *Let $a_1, \dots, a_k$ be relatively prime positive integers with $a_1 \leq a_i$ for all i , and n an integer $\geq n_0 = (a_1 - 1)(a_2 + \dots + a_k - 1)$. Then there are nonnegative integers $x_1, \dots, x_k$ such that $n = a_1x_1 + \dots + a_kx_k$.*

Lemma 5. *Let $a_1, \dots, a_k$ be relatively prime positive integers and n an arbitrary integer. Then there are integers $x_1, \dots, x_k$ such that $n = a_1x_1 + \dots + a_kx_k$ and $0 \leq x_i < a_1$ for $i \geq 2$.*

Proof. There are integers $y_1, \dots, y_k$ such that $n = a_1y_1 + \dots + a_ky_k$. Writing $\bar{u}$ for the congruence class mod a_1 of the integer u , we get $\bar{n} = \bar{a_2y_2} + \dots + \bar{a_ky_k}$. For $i \geq 2$ let $x_i \in \mathbb{Z}$ satisfy $\bar{x_i} = \bar{y_i}$ and $0 \leq x_i < a_1$. Since $\bar{x_i} = \bar{y_i}$ for $i \geq 2$, we have $\bar{n} = \bar{a_2x_2} + \dots + \bar{a_kx_k}$, which means that $n - a_2x_2 - \dots - a_kx_k$ is a multiple of a_1 . Therefore, there is an integer x_1 such that $n - a_2x_2 - \dots - a_kx_k = a_1x_1$, and the integers $x_1, \dots, x_k$ do the job. $\square$

Proof of Proposition 4. Let n and $x_1, \dots, x_k$ be as in Lemma 5. Since $x_2, \dots, x_k$ are nonnegative, it is enough to show $x_1 \geq 0$ if $n \geq n_0$ (where n_0 is as in the statement of the proposition). We have

$$n = a_1x_1 + \sum_{i=2}^k a_ix_i \leq a_1x_1 + \sum_{i=2}^k a_i(a_1 - 1) = a_1x_1 + m,$$

with $m = (a_1 - 1)(a_2 + \dots + a_k)$, hence $x_1 \geq (n - m)/a_1$, and we see that $(n - m)/a_1 > -1$ implies $x_1 \geq 0$. We have

$$\frac{n - m}{a_1} > -1 \Leftrightarrow n > m - a_1 \Leftrightarrow n \geq m - a_1 + 1 = n_0,$$

so $n \geq n_0$ implies $x_1 \geq 0$, as required. $\square$

References

- [1] Jean Gallier. *The Frobenius Coin Problem Upper Bounds on The Frobenius Number*. <https://www.cis.upenn.edu/~cis5110/Frobenius-number.pdf> or [https://github.com/Pierre-Yves-Gaillard/weil-beginners-exercises/blob/main/Frobenius-number-Gaillard.pdf](https://github.com/Pierre-Yves-Gaillard/weil-beginners-exercises/blob/main/Frobenius-number-Gallier.pdf). 2014.
- [2] André Weil. *Number Theory for Beginners*. Springer, 1979.